

Polymer Simulation Knowledge Base

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Chapter 1

Polymer Simulation

A practical knowledge base for molecular simulation of polymers. Find literature, workflows, software notes, and reusable scripts for building and analyzing polymer models.

[PDF build workflow](#)

From atomistic models to coarse-grained melts: a growing, source-linked reference for reproducible polymer simulation.

1.1 Start here

I want to...	Go to
Find foundational papers and a suggested reading path	Literature
Learn a simulation workflow	Tutorials
Compare simulation packages and parameterization tools	Tools
Find or contribute scripts and reproducible examples	Scripts
Browse datasets, templates, and external references	Resources
Read the complete guide as a book	Download the PDF

1.2 What's covered

- **Models:** atomistic, united-atom, and coarse-grained polymer systems
- **Methods:** force-field selection, parameterization, equilibration, and multiscale modeling
- **Applications:** melts, glassy polymers, electrolytes, and nanocomposites
- **Tools:** LAMMPS, GROMACS, RadonPy, and related packages
- **Analysis:** chain statistics, dynamics, entanglement, and material properties

Literature-reported MD results can differ from uniformly generated computational datasets because outcomes depend on the force field, model representation, polymer chemistry, and simulation protocol.

1.3 Repository map

Directory	Contents
papers/	Reading lists, literature notes, and citation collections
tutorials/	Step-by-step workflows and simulation recipes
scripts/	Reusable scripts, prototypes, and reproducible examples
tools/	Software, utilities, installation, and configuration notes
resources/	Links, datasets, benchmarks, and templates

1.4 Build the PDF

The PDF combines the Markdown guides into one document. Install [Pandoc](#) and XeLaTeX, then run:

```
make
```

See `make help` for available targets. The PDF is also built automatically on pushes and pull requests to `main`.

1.5 Contributing

Add material to the directory that best fits its purpose. Keep notes concise, link claims to their sources, and include dependencies and example usage with scripts. When adding a new Markdown chapter, add it to `SOURCES` in the `Makefile` so it appears in the PDF.

Chapter 2

Papers

Use this folder for literature notes, reading lists, citation collections, and short paper summaries related to polymer simulation.

2.1 VIP Paper

1. High-Throughput Molecular Dynamics Simulations and Validation of Thermophysical Properties of Polymers for Various Applications, ACS Appl. Polym. Mater. (2021) 3 (2): 620–630., <https://doi.org/10.1021/acsapm.0c00524>
2. Paudel, A., Shakya, A., Nag, S. et al. A structured dataset of polymers in molecular dynamics simulations extracted by LLM-assisted text mining. Sci Data (2026). <https://doi.org/10.1038/s41597-026-08045-x>
3. Macromolecules (2019) 52 (3): 755–786. <https://doi.org/10.1021/acs.macromol.8b01836>, Modeling and Simulations of Polymers: A Roadmap
4. Chew, A.K., Afzal, M.A.F., Kaplan, Z. et al. Leveraging high-throughput molecular simulations and machine learning for the design of chemical mixtures. npj Comput Mater 11, 72 (2025). <https://doi.org/10.1038/s41524-025-01552-2>
5. Atomistic polymer modeling: Recent advances and challenges in building and parametrization workflows HN Turney, M Matta - Macromolecules, 2025

2.2 Benchmark

1. Benchmarking Classical Molecular Dynamics Simulations for Computational Screening of Lithium Polymer Electrolytes, Macromolecules (2025) 58 (13): 6732–6742. <https://doi.org/10.1021/acs.macromol.5c00930>

2.3 High-Citation and Classic Reading List

Citation counts are approximate screening signals, because Google Scholar, ACS, Scopus, Crossref, OUCI, and other databases use different coverage. Counts below were checked from publicly visible pages on 2026-05-20.

Year	Paper	Why it matters	Citation signal
1985	Theodorou & Suter, <i>Detailed molecular structure of a vinyl polymer glass</i>	Early atomistic modeling of amorphous polymer glass structure; useful for packing, free volume, and glassy polymer model-building ideas.	ACS: cited by ~823 publications.
1990	Kremer & Grest, <i>Dynamics of entangled linear polymer melts: A molecular-dynamics simulation</i>	The foundational bead-spring / Kremer-Grest polymer melt model; essential for entanglement, Rouse-to-reptation crossover, and generic polymer MD.	OUCI: ~3,572 citations.
1990	Mayo, Olafson & Goddard, <i>DREIDING: A generic force field for molecular simulations</i>	General-purpose force field often encountered in polymer, organic, and materials simulations. Not polymer-specific, but important background for force-field choices.	ACS: cited by ~5,948 publications.
1997	Paul, Smith & Yoon, <i>Static and Dynamic Properties of a n-C100H202 Melt from Molecular Dynamics Simulations</i>	Representative atomistic/united-atom polyethylene melt simulation; good example of comparing MD chain statistics and dynamics with experiment.	Classic atomistic PE melt paper.
1998	Sun, <i>COMPASS: An Ab Initio Force-Field Optimized for Condensed-Phase Applications</i>	Widely used all-atom force field for condensed-phase simulations of organic molecules and polymers; common in Materials Studio/Forcite workflows.	CoLab: ~5,474 citations.

Year	Paper	Why it matters	Citation signal
1998	Bennemann, Paul, Binder & Dunweg, <i>Molecular-dynamics simulations of the thermal glass transition in polymer melts</i>	Classic dense polymer melt glass-transition MD study; useful for alpha relaxation, cooling protocols, and mode-coupling-theory comparisons.	Classic glass-transition simulation paper.
2000	Borodin & Smith, <i>Molecular Dynamics Simulations of Poly(ethylene oxide)/LiI Melts. 2. Dynamic Properties</i>	Important polymer-electrolyte MD paper; useful for ion transport, segmental dynamics, and PEO-salt coordination.	ACS: cited by ~115 publications.
2001	Akkermans & Briels, <i>A structure-based coarse-grained model for polymer melts</i>	A key bottom-up coarse-graining paper; highlights structure reproduction versus thermodynamic consistency.	University of Twente/Scopus: ~125 citations.
2002	Muller-Plathe, <i>Coarse-graining in polymer simulation: From the atomistic to the mesoscopic scale and back</i>	Review-style entry point for systematic polymer coarse graining and backmapping.	Classic coarse-graining review.
2003	Reith, Putz & Muller-Plathe, <i>Deriving effective mesoscale potentials from atomistic simulations</i>	Canonical iterative Boltzmann inversion style reference for deriving coarse-grained potentials from atomistic data.	Widely used coarse-graining method paper.
2003	Smith et al., <i>A molecular dynamics simulation study of nanoparticle interactions in a model polymer-nanoparticle composite</i>	Early MD study of polymer-mediated nanoparticle interactions in a polymer matrix; useful for polymer nanocomposites.	ScienceDirect: cited by ~262 publications.

Year	Paper	Why it matters	Citation signal
2003	Starr, Douglas & Glotzer, <i>Origin of particle clustering in a simulated polymer nanocomposite and its impact on rheology</i>	Classic simulation paper on nanoparticle clustering, dispersion, shear, and rheology in polymer nanocomposites.	NIST-hosted publication; widely cited in PNC literature.
2004	Everaers et al., <i>Rheology and Microscopic Topology of Entangled Polymeric Liquids</i>	Primitive-path analysis foundation linking microscopic entanglement topology to rheology and plateau modulus.	Seminal Science paper.
2004	Faller, <i>Automatic coarse graining of polymers</i>	Clear review of automatic mapping and potential-optimization schemes for polymer coarse graining.	ScienceDirect: cited by ~102 publications.
2004	Paul & Smith, <i>Structure and dynamics of amorphous polymers: computer simulations compared to experiment and theory</i>	Broad review connecting atomistic and coarse-grained simulations with amorphous polymer experiments.	Broad review paper.
2006	Harmandaris et al., <i>Hierarchical Modeling of Polystyrene: From Atomistic to Coarse-Grained Simulations</i>	Practical multiscale workflow for polystyrene: atomistic model, coarse-graining, time mapping, and backmapping.	ACS: cited by ~334 publications.
2014	Liu et al., <i>Molecular dynamics simulation of dispersion and aggregation kinetics of nanorods in polymer nanocomposites</i>	Useful example for rod-like fillers, dispersion/aggregation kinetics, and CG polymer nanocomposite analysis.	ScienceDirect: cited by ~45 publications.
2017	Vogiatzis & Theodorou, <i>Multiscale Molecular Simulations of Polymer-Matrix Nanocomposites</i>	Open-access review of polymer-matrix nanocomposite simulations across length scales.	Springer: ~69 citations.

Year	Paper	Why it matters	Citation signal
2023	Shi, Qian & Lu, <i>Coarse-grained molecular dynamics simulation of polymers: Structures and dynamics</i>	Recent high-level review for CG polymer MD, including structure, dynamics, and chemically specific coarse-grained models.	Recent WIREs review.

2.4 Suggested Reading Path

1. Start with the generic melt baseline: Kremer & Grest 1990.
2. Add atomistic polymer examples: Theodorou & Suter 1985; Paul, Smith & Yoon 1997.
3. Learn force-field context: DREIDING 1990; COMPASS 1998.
4. Study glassy dynamics: Bennemann et al. 1998; Paul & Smith 2004.
5. Learn coarse graining: Akkermans & Briels 2001; Muller-Plathe 2002; Reith et al. 2003; Faller 2004; Harmandaris et al. 2006; Shi et al. 2023.
6. Study entanglement topology and rheology: Everaers et al. 2004.
7. Move to applications: Borodin & Smith 2000 for polymer electrolytes; Smith et al. 2003 and Starr et al. 2003 for polymer nanocomposites.

2.5 Tags to Use Later

- polymer-melt
- kremer-grest
- atomistic-md
- united-atom
- force-field
- coarse-graining
- backmapping
- glass-transition
- entanglement
- rheology
- polymer-electrolyte
- polymer-nanocomposite

2.6 To Expand

- Add BibTeX entries in papers/bib/.
- Add one Markdown note per paper in papers/notes/.
- Add a separate list for polymer crystallization, cross-linked polymers/epoxy, polymer membranes, and LAMMPS/RadonPy-specific workflows.

Chapter 3

Tutorials

Use this folder for step-by-step workflows, setup notes, simulation recipes, and learning materials.

Chapter 4

Scripts

Use this folder for reusable scripts, small prototypes, and reproducible examples for polymer simulation.

Organize material by workflow where useful: system building, simulation setup, trajectory analysis, plotting, and automation. Include dependencies and a short example of how to run each script.

Chapter 5

Tools

Use this folder for notes about simulation software, analysis packages, utilities, installation steps, and configuration tips.

5.1 Classical Force Fields

1. Amber1
2. CHARMM2
3. OPLS
4. PCFF4
5. GROMOS
6. COMPASS
7. United Atoms (UA)
8. Coarsed-Grained (GC)
9. ReaxFF

5.2 machine-learning force fields (MLFFs)

- SimPoly
- n-the-Fly Machine-Learned Force Fields for High-Fidelity Polymer Glass Transition Simulations JCIM, 2026
- How long is enough? JCTC, 2026
- Active Learning of a Neural Network Potential for Large-Scale Atomistic Simulations of Polymer Electrolyte Membranes, J. Phys. Chem. B (2026) 130 (36): 9326–9337. <https://doi.org/10.1021/acs.jpcc.6c04744>

5.3 Simulation Software

1. Gromacs
2. Lammmps
3. Material Studio
4. [RadonPy](#)
5. pysimm

6. [SMiPoly](#)
7. SPACIER
8. [PEMD](#)
9. [PySoftK](#)
10. Polymer-Agent (2026)
11. ToPolyAgent (2026)
12. [MolPy](#)
13. [SimPoly](#) (2025)
14. NAMD
15. CHARMM
16. MARTINI
17. PolyRapid: automated high-throughput screening of polymers using a computational workflow Aug 26 2026 SOFT MATTER
18. Glass Transition Prediction of Binary Copolymers Across Large Chemical Spaces Using Machine Learning and Physics-Based Modeling Jul 14 2026 POLYMERS
19. How Long Is Long Enough? Extrapolation of Machine-Learning Interatomic Potentials for Oligomeric and Polymeric Systems Jul 14 2026 JOURNAL OF CHEMICAL THEORY AND COMPUTATION
20. From oligomers to entangled polymers: How to train a transferable machine learning interatomic potential
21. ADEPT-PolyGraphMT: automated molecular simulation and multi-task multi-fidelity machine learning for polymer property generation and prediction Jul 1 2026 DIGITAL DISCOVERY
22. Atomistic Polymer Modeling: Recent Advances and Challenges in Building and Parametrization Workflows Nov 11 2025 MACROMOLECULES
23. Dissipative Particle Dynamics Modeling in Polymer Science and Engineering Apr 15 2025 WILEY INTERDISCIPLINARY REVIEWS-COMPUTATIONAL MOLECULAR SCIENCE
24. Rapid, Accurate and Reproducible Prediction of the Glass Transition Temperature Using Ensemble-Based Molecular Dynamics Simulation Jan 29 2025 JOURNAL OF CHEMICAL THEORY AND COMPUTATION
25. SPACIER: on-demand polymer design with fully automated all-atom classical molecular dynamics integrated into machine learning pipelines Jan 28 2025 NPJ COMPUTATIONAL MATERIALS
26. PolyGraphPy, 2026, computational material science.

5.4 Force Field Parameterization

1. Soptop
2. LigPargen
3. [PolyPargen](#)

5.5 Automatic AI Agent Workflow for molecular dynamic simulation

1. [G-BigSMILES 2.0](#)
2. [PolyJarvis](#)
3. [MDCROW](#)

4. Polymer-Agent

Chapter 6

Resources

Use this folder for useful links, datasets, templates, benchmark inputs, and miscellaneous references.

6.1 MD data collection

1. <https://polydatabase.com/>
2. SimPoly
3. Poly Omics
4. OPoly26